

## WOLFSEGG, Austria

WMO: 110010

Lat: 48.106N

Lon: 13.671E

Elev: 616

StdP: 94.14

Time Zone: 1.00 (EUC)

Period: 97-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -10.5      | -8.7 | -13.2                         | 1.3 | -9.5 | -11.0 | 1.6 | -7.8 | 14.4                  | 5.6  | 12.5 | 3.7  | 3.0                   | 80   | 0.518 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 7.1                    | 28.1            | 19.6 | 26.2 | 18.9 | 24.7 | 18.1 | 20.3                | 26.5 | 19.4 | 25.2 | 18.6 | 23.8 | 2.2                  | 100  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 18.0                            | 14.0 | 23.5 | 17.1 | 13.2 | 22.4 | 16.5 | 12.6 | 21.6 | 61.0          | 26.7 | 58.0 | 25.4 | 55.1 | 24.0 | 24.0              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 10.6          | 8.9  | 7.5 | -12.2                      | 30.5 | 2.9                | 1.6 | -14.3                                              | 31.6 | -16.0      | 32.5 | -17.6      | 33.4 | -19.7      | 34.5 |    |    |
| (5)               |      |     | -12.9                      | 22.0 | 2.5                | 1.2 | -14.7                                              | 22.9 | -16.2      | 23.6 | -17.6      | 24.3 | -19.5      | 25.2 |    |    |

## Monthly Climatic Design Conditions

|      |                                                                                  | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |      |
|------|----------------------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|------|
|      |                                                                                  | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |      |
| (6)  | Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 8.9   | -1.4  | 0.6   | 4.3   | 9.2   | 13.0  | 16.6  | 18.4  | 18.1  | 13.7  | 9.0   | 3.6   | 0.5  | (6)  |
| (7)  |                                                                                  | DBStd       | 8.07  | 4.73  | 4.91  | 4.69  | 4.43  | 4.06  | 4.19  | 3.71  | 3.57  | 3.55  | 4.20  | 4.09  | 4.27 | (7)  |
| (8)  |                                                                                  | HDD10.0     | 1451  | 352   | 264   | 183   | 67    | 17    | 3     | 0     | 0     | 7     | 69    | 196   | 294  | (8)  |
| (9)  |                                                                                  | HDD18.3     | 3588  | 611   | 497   | 434   | 274   | 170   | 82    | 46    | 49    | 144   | 288   | 442   | 552  | (9)  |
| (10) |                                                                                  | CDD10.0     | 1034  | 0     | 1     | 7     | 42    | 111   | 201   | 260   | 251   | 118   | 39    | 4     | 0    | (10) |
| (11) |                                                                                  | CDD18.3     | 129   | 0     | 0     | 0     | 1     | 6     | 29    | 48    | 41    | 5     | 0     | 0     | 0    | (11) |
| (12) |                                                                                  | CDH23.3     | 666   | 0     | 0     | 0     | 2     | 20    | 161   | 258   | 213   | 12    | 0     | 0     | 0    | (12) |
| (13) |                                                                                  | CDH26.7     | 120   | 0     | 0     | 0     | 0     | 1     | 27    | 47    | 45    | 1     | 0     | 0     | 0    | (13) |
| (14) | Wind                                                                             | WSAvg       | 3.2   | 4.2   | 4.2   | 4.1   | 3.3   | 3.0   | 2.5   | 2.6   | 2.6   | 2.7   | 3.0   | 3.3   | 3.5  | (14) |
| (15) | Precipitation                                                                    | PrecAvg     | 1025  | 63    | 58    | 76    | 64    | 105   | 126   | 126   | 119   | 88    | 70    | 64    | 67   | (15) |
| (16) |                                                                                  | PrecMax     | 1299  | 126   | 130   | 140   | 161   | 201   | 298   | 267   | 304   | 186   | 137   | 154   | 121  | (16) |
| (17) |                                                                                  | PrecMin     | 734   | 11    | 11    | 8     | 6     | 31    | 50    | 29    | 34    | 25    | 16    | 0     | 11   | (17) |
| (18) |                                                                                  | PrecStd     | 125   | 35    | 31    | 40    | 36    | 44    | 50    | 51    | 49    | 38    | 31    | 33    | 31   | (18) |
| (19) | Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB    | 11.2  | 13.6  | 16.8  | 22.3  | 25.6  | 29.3  | 30.2  | 30.7  | 24.7  | 20.4  | 16.0  | 11.7 | (19) |
| (20) |                                                                                  |             | MCWB  | 7.0   | 7.4   | 10.2  | 14.0  | 17.8  | 20.6  | 20.4  | 19.8  | 18.1  | 14.6  | 11.1  | 7.6  | (20) |
| (21) |                                                                                  | 2%          | DB    | 9.0   | 10.7  | 14.8  | 19.7  | 23.5  | 27.1  | 27.9  | 27.7  | 22.6  | 18.1  | 13.3  | 9.6  | (21) |
| (22) |                                                                                  |             | MCWB  | 6.2   | 7.0   | 9.1   | 12.8  | 16.6  | 19.5  | 19.4  | 19.3  | 17.4  | 13.7  | 9.6   | 6.4  | (22) |
| (23) |                                                                                  | 5%          | DB    | 7.1   | 8.9   | 12.8  | 17.7  | 21.6  | 25.1  | 26.0  | 25.5  | 20.8  | 16.2  | 11.3  | 8.1  | (23) |
| (24) |                                                                                  |             | MCWB  | 4.7   | 5.8   | 8.1   | 11.5  | 15.5  | 18.2  | 18.8  | 18.7  | 16.3  | 12.8  | 8.8   | 5.5  | (24) |
| (25) |                                                                                  | 10%         | DB    | 5.2   | 7.2   | 10.9  | 15.9  | 19.6  | 23.3  | 24.5  | 23.9  | 19.1  | 14.7  | 9.3   | 6.5  | (25) |
| (26) |                                                                                  |             | MCWB  | 3.4   | 4.7   | 7.1   | 10.6  | 14.2  | 17.3  | 18.1  | 17.9  | 15.4  | 12.0  | 7.6   | 4.5  | (26) |
| (27) | Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB    | 8.1   | 9.0   | 11.0  | 14.7  | 18.9  | 21.7  | 21.4  | 21.2  | 19.1  | 15.7  | 11.6  | 8.3  | (27) |
| (28) |                                                                                  |             | MCDWB | 10.1  | 11.9  | 15.7  | 20.9  | 24.3  | 27.6  | 27.8  | 27.9  | 23.4  | 19.1  | 15.0  | 10.4 | (28) |
| (29) |                                                                                  | 2%          | WB    | 6.4   | 7.3   | 9.5   | 13.2  | 17.1  | 20.1  | 20.1  | 20.0  | 17.8  | 14.3  | 10.1  | 7.1  | (29) |
| (30) |                                                                                  |             | MCDWB | 8.6   | 10.1  | 13.7  | 19.0  | 22.6  | 26.3  | 26.4  | 26.1  | 22.2  | 17.0  | 12.5  | 9.1  | (30) |
| (31) |                                                                                  | 5%          | WB    | 5.1   | 6.0   | 8.5   | 12.0  | 15.8  | 18.8  | 19.3  | 19.1  | 16.7  | 13.2  | 8.9   | 5.9  | (31) |
| (32) |                                                                                  |             | MCDWB | 6.9   | 8.6   | 12.1  | 17.1  | 20.8  | 24.4  | 25.2  | 24.6  | 20.1  | 15.6  | 11.0  | 7.7  | (32) |
| (33) |                                                                                  | 10%         | WB    | 3.6   | 4.8   | 7.4   | 10.8  | 14.7  | 17.7  | 18.4  | 18.2  | 15.7  | 12.3  | 7.6   | 4.6  | (33) |
| (34) |                                                                                  |             | MCDWB | 5.1   | 7.0   | 10.4  | 15.2  | 19.1  | 22.5  | 23.6  | 23.2  | 18.7  | 14.4  | 9.3   | 6.3  | (34) |
| (35) | Mean Daily<br>Temperature<br>Range                                               | MDBR        | 3.7   | 4.4   | 5.8   | 7.0   | 7.2   | 7.0   | 7.1   | 6.9   | 5.9   | 5.1   | 3.8   | 3.9   | (35) |      |
| (36) |                                                                                  | 5% DB       | MCDBR | 5.7   | 6.5   | 8.4   | 9.2   | 9.6   | 9.5   | 9.1   | 9.2   | 7.8   | 6.9   | 6.6   | 6.3  | (36) |
| (37) |                                                                                  |             | MCWBR | 3.9   | 4.3   | 5.1   | 5.1   | 5.3   | 4.9   | 4.3   | 4.4   | 4.4   | 4.4   | 4.4   | 4.5  | (37) |
| (38) |                                                                                  |             | 5% WB | MCDBR | 5.5   | 6.1   | 7.7   | 8.9   | 9.1   | 8.9   | 8.7   | 8.7   | 7.3   | 6.2   | 5.9  | 6.1  |
| (39) |                                                                                  | MCWBR       |       | 4.1   | 4.3   | 5.0   | 5.1   | 5.3   | 4.9   | 4.4   | 4.5   | 4.3   | 4.3   | 4.2   | 4.6  | (39) |
| (40) | Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.288 | 0.312 | 0.350 | 0.381 | 0.390 | 0.394 | 0.400 | 0.393 | 0.368 | 0.339 | 0.313 | 0.287 | (40) |      |
| (41) |                                                                                  | taud        | 2.423 | 2.366 | 2.334 | 2.297 | 2.306 | 2.328 | 2.308 | 2.345 | 2.393 | 2.462 | 2.487 | 2.461 | (41) |      |
| (42) |                                                                                  | Ebn at Noon | 793   | 846   | 864   | 868   | 872   | 869   | 857   | 847   | 835   | 806   | 756   | 749   | (42) |      |
| (43) |                                                                                  | Edh at Noon | 66    | 87    | 105   | 121   | 124   | 123   | 123   | 114   | 99    | 79    | 62    | 56    | (43) |      |
| (44) | All-Sky Solar<br>Radiation                                                       | RadAvg      | 1.11  | 2.03  | 3.06  | 4.23  | 4.89  | 5.39  | 5.29  | 4.68  | 3.37  | 2.21  | 1.17  | 0.87  | (44) |      |
| (45) |                                                                                  | RadStd      | 0.16  | 0.27  | 0.37  | 0.53  | 0.58  | 0.50  | 0.51  | 0.51  | 0.45  | 0.24  | 0.16  | 0.12  | (45) |      |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |  | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (99 neighbors) |  | +0.48 | N/A     | N/A    | N/A     | +0.40 | +0.36 | N/A         | N/A     | +68     | +30     |

Nomenclature: See separate page